

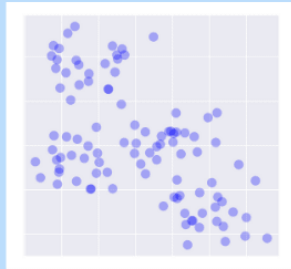
K-MEANS ALGORITHM

ALGORITHM OVERVIEW

Example: A Bank wants to segment its customers into 4 groups

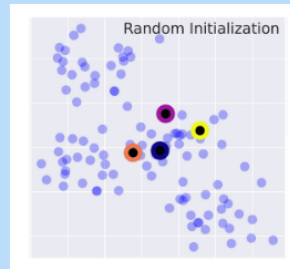
Step 1

Select $K = 4$ as we need 4 groups



Step 2

Randomly select 4 distinct data points. These are called cluster centroids.



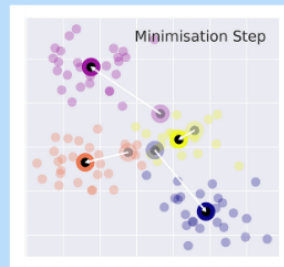
Step 3

Expectation: Assign every point to the closest cluster centroid.



Step 4

Minimisation: Move the cluster centroid to the average of the points in a cluster



Repeat

Repeat steps 3 and 4 until there are no changes in clusters



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Code adapted from the Python Data Science Handbook by Jake VanderPlas

PRACTICE WITH THE K-MEANS ALGORITHM

Exercise 01:

Given a dataset of 8 students, where each point (x, y) represents a student's age and marks, respectively:

$S_1(5,10)$, $S_2(6,8)$, $S_3(4,5)$, $S_4(7,10)$, $S_5(8,12)$, $S_6(10,9)$, $S_7(12,11)$, and $S_8(4,6)$.

The initial cluster centers are: $\mu_1 = S_1(5,10)$, $\mu_2 = S_4(7,10)$, and $\mu_3 = S_7(12,11)$.

- Perform the clustering using the Manhattan distance metric.
- Perform the clustering using the Euclidean distance metric.
- Compare the number of iterations required to reach convergence between the two distance metrics.
- Implement the algorithm programmatically and visualize the results at each iteration. For each iteration, display two plots (side-by-side or sequentially):
 - The clustering result** showing data points colored by their assigned cluster and the current cluster centroids.
 - The centroid shift**, visualizing the trajectory of each centroid from its previous position to its updated position.

Note: For manual calculations in parts (a) and (b), record your results at each iteration using the following table format:

			(5, 10)	(7, 10)	(12, 11)	
No	Age	Marks	Dist Mean 1	Dist Mean 2	Dist Mean 3	Cluster

FUZZY C-MEANS ALGORITHM

FCM

Require: Set values for the number of clusters C , the degree of fuzziness $m > 1$ and the error ϵ .

1: Initialize randomly the centers of clusters $c_i^{(0)}$.

2: $k \leftarrow 1$

3: **repeat**

4: Calculate the membership matrix $U^{(k)}$ using the centers $c_i^{(k-1)}$:

$$u_{ij}^{(k)} \leftarrow \frac{1}{\sum_{l=1}^C \left(\frac{d(x_j, c_l^{(k-1)})}{d(x_j, c_i^{(k-1)})} \right)^{\frac{2}{m-1}}}$$

5: Update the centers $c_i^{(k)}$ using $U^{(k)}$: $c_i^{(k)} \leftarrow \frac{\sum_{j=1}^N (u_{ij}^{(k)})^m x_j}{\sum_{j=1}^N (u_{ij}^{(k)})^m}$

6: $k \leftarrow k + 1$

7: **until** $\|c_i^{(k)} - c_i^{(k-1)}\| < \epsilon$

8: **return** c_i the centers of clusters and the membership degrees u_{ij}

$U = [u_{ij}]_{C \times N}$ is the membership function matrix,
 $d(x_j, c_i)$ is the metric which calculates the distance between the element x_j and the center of cluster c_i ,
 C is the number of clusters,
 N is the number of data,
 m is the degree of fuzziness ($m > 1$).

PRACTICE WITH THE FUZZY C-MEANS ALGORITHM

Exercise 02:

Given a dataset of 8 students, where each point (x, y) represents a student's age and marks, respectively:

$S_1(5,10)$, $S_2(6,8)$, $S_3(4,5)$, $S_4(7,10)$, $S_5(8,12)$, $S_6(10,9)$, $S_7(12,11)$, and $S_8(4,6)$.

The initial cluster centers are: $C_1 = S_1(5,10)$, $C_2 = S_4(7,10)$, and $C_3 = S_7(12,11)$.

Suppose we want to cluster this dataset into $C = 3$ fuzzy clusters with the fuzzification parameter $m = 2$.

a. Perform the Fuzzy C-Means algorithm step-by-step for the first 3 iterations. For each iteration, present your calculations using the two standard tables below:

- Distance & Membership Computation:

No	Age	Marks	Dist to C1	Dist to C2	Dist to C3	μ_1 (Cluster 1)	μ_2 (Cluster 2)	μ_3 (Cluster 3)	$\sum \mu$

- Cluster Centroid Update:

Centroid	Old Coordinates	Updated Coordinates

b. Implement the Fuzzy C-Means algorithm programmatically with a convergence tolerance $\epsilon = 10^{-1}$ (or a maximum of 20 iterations). For each iteration, display two plots:

- **The clustering result**, showing data points with soft cluster memberships (using a continuous colormap representing μ) and the current centroids.

- **The centroid shift**, visualizing the displacement/movement vector of each centroid from its previous position to its updated position.

c. Using the program developed in **Exercise 01 and 02**, run both K-Means and Fuzzy C-Means on this dataset. Compare their performance and outcomes regarding:

- The boundary points and soft assignments versus hard cluster boundaries.
- Convergence rate (number of iterations required).
- Final cluster center coordinates.